

VERITAS / CAGE Phase 4 — Invariant Review Specification

Provider03 Boundary Invariant Review Before Runtime Prototype

Revision Status

Phase: Phase 4 — Invariant Review

Status: DRAFT — Ready for CAGE-side review

VERITAS baseline: veritas_os/main @ ee3cc6c4d43fea76aa62daebdb466757ee3bad80

CAGE Phase 3 contract reference: ecbcce3253cb55c333f47009c40a45dbc5b3a454

Phase 3: COMPLETE / CAGE-side reviewed and approved

Phase 5 Runtime Prototype: NOT STARTED

1. Purpose

This specification defines the invariants that must remain true across the VERITAS / CAGE Provider03 interoperability boundary before any Phase 5 runtime prototype wiring begins.

Phase 3 demonstrated a deterministic, synthetic, side-effect-free interoperability proof over the existing seven VERITAS AML/KYC regulated-action scenarios. Phase 4 does not add live execution. Instead, it identifies and reviews the properties that must never be weakened by later adapter, runtime, evidence, transport, or integration changes.

The core question is:

What must remain true even when implementation details change?

2. Non-Goals

Phase 4 does not establish or claim:

- live CAGE runtime integration;
- production deployment;
- live customer credentials or endpoints;
- regulatory approval or certification;
- commercial integration.

3. Source Baselines

VERITAS: veritas_os/main @ ee3cc6c4d43fea76aa62daebdb466757ee3bad80

CAGE reference: ecbcce3253cb55c333f47009c40a45dbc5b3a454

Provider03: src/integrations/provider_03/provider.py

Phase 3 status: COMPLETE / CAGE-side reviewed and approved.

4. Review Model

Each invariant is reviewed as: invariant, normative rule, failure condition, and evidence.

5. Invariant Set

I-01 — AI decision or evidence does not create execution authority

MUST keep decision/evidence separate from Authority. MUST NOT turn an AI recommendation, APPROVED verdict, or evidence hash into permission to execute.

I-02 — Provider03 does not amplify authority or admissibility

MUST preserve or narrow actor/action/target/scope state. MUST NOT widen authority, credential scope, policy scope, or approval scope.

I-03 — Fail-closed behavior is monotonic

Missing, unknown, stale, malformed, mismatched, unverified, or transport-error states MUST remain non-admitted. No fallback-to-allow path is permitted.

I-04 — ESCALATE is never equivalent to APPROVED

MUST preserve ESCALATE → admitted=false + needs_human_review=true. Human-review routing is not temporary execution permission.

I-05 — Authority, Policy, and Human Approval remain separate evidence domains

Each MUST be verified independently. Identity must not substitute for authority; review-required state must not substitute for approval.

I-06 — Action, target, resource, and scope cannot widen across the boundary

Exact governed action and target bindings MUST be preserved. No implicit wildcard, alias, hierarchy, or scope expansion is allowed.

I-07 — action_context mapping is explicit and non-mutating

Current mapping is amount → magnitude and symbol → context. Mapping MUST be caller-declared and MUST NOT mutate caller state.

Open Phase 4 question: when both a legacy key and its canonical destination key are present, the collision rule must be explicit before Phase 5. Silent overwrite must not be assumed.

I-08 — VERITAS bind-receipt hash and CAGE ingest digest remain separate hash domains

Names, inputs, and canonicalization rules MUST remain distinct. Equality must never be assumed unless the same canonical bytes are intentionally hashed.

I-09 — Evidence hash proves integrity, not authority or signature

A SHA-256/JCS digest is an integrity identifier only. It MUST NOT be treated as issuer identity, approval, authority, or signature proof.

I-10 — TrustLog lineage cannot be promoted into a witness claim

When no real witness-generation path ran, trustlog_proof_status MUST remain lineage_only and witness_hash MUST remain absent/null.

I-11 — Determinism is preserved for the bounded proof domain

Identical canonical inputs MUST produce identical governance-significant proof outputs. Non-deterministic timestamps, random identifiers, unstable ordering, or unsupported number handling must not enter proof identity.

I-12 — BindReceipt and Outcome remain retrospective evidence

Receipt existence MUST NOT grant new authority, refresh expired authority, or bypass current rechecks.

I-13 — CAGE routing/enforcement seal and VERITAS BindReceipt remain distinct controls

They MUST retain separate purposes, key/control domains, lifecycles, and verification semantics. Neither may substitute for the other.

I-14 — CAGE integration cannot silently modify the frozen VERITAS Decision-to-Effect chain

Phase 5 MUST preserve authorization issuance $\neq$ execution permission, single-use consumption, fresh governance/authority/approval/risk rechecks, durable dispatch intent, EFFECT_UNKNOWN semantics, no blind redispach, independent reconciliation, and no authority creation from BindReceipt/Outcome.

I-15 — External UTC Clock Evidence remains an independent evidence boundary

VerifiedExternalClockEvidence is time evidence only. It MUST NOT become AuthorityEvidence, HumanApproval, BindAuthorization, execution permission, or implicit CAGE admission without a separate reviewed composition contract.

I-16 — Reviewer/proof artifacts are audit evidence, not portable runtime trust

Source-bound CI artifacts may support replay and review, but MUST NOT be deserialized and silently treated as current trusted runtime state.

6. Invariant Verification Matrix

- I-01 Authority separation — Required
- I-02 No authority amplification — Required
- I-03 Fail-closed monotonicity — Required
- I-04 ESCALATE remains non-admitted — Required
- I-05 Authority/Policy/Approval separation — Required
- I-06 Exact action/target/scope binding — Required
- I-07 action_context mapping — OPEN until collision behavior is fixed
- I-08 Hash-domain separation — Required
- I-09 Digest is integrity-only — Required
- I-10 TrustLog lineage_only integrity — Required
- I-11 Deterministic replay — Required
- I-12 Receipt/Outcome remain evidentiary — Required
- I-13 Routing seal / BindReceipt separation — Required
- I-14 Frozen execution semantics preserved — Required

I-15 External UTC clock boundary remains separate — Required

I-16 Reviewer artifact is not runtime trust — Required

7. Required Phase 4 Review Questions

1. Does CAGE agree that these invariants are valid entry constraints for Phase 5?
2. What is the normative collision behavior when both legacy and canonical action_context keys are present?
3. Should mapping collisions be rejected before upstream dispatch or surfaced with a deterministic finding code?
4. Are additional DeferQueue, routing-seal, or EvidenceSeal invariants required before runtime wiring?
5. Should the invariant set also be represented as a machine-readable conformance artifact?

8. Fail-Closed Review Cases

Phase 4 requires explicit confirmation that unknown or missing verdicts, endpoint or transport failures, missing or expired authority, missing required approval, stale evidence, policy uncertainty, mapping collision, target mismatch, tampered receipt/evidence, unsupported TrustLog witness claims, and reviewer-artifact replay cannot become admitted by fallback behavior.

9. Relationship to Phase 3

Phase 3 remains the deterministic implementation-proof baseline. Phase 4 does not rewrite the seven AML/KYC scenarios or the approved 1 / 1 / 5 verdict distribution. It converts demonstrated properties into explicit constraints for runtime composition.

10. Relationship to the Frozen VERITAS Execution Architecture

The Controlled Execution Proof Architecture Freeze v1 remains authoritative for the proven Decision-to-Effect sandbox chain. If CAGE runtime wiring would change authorization, single-use consumption, effect uncertainty, credential or network authority, reconciliation, receipt/outcome semantics, or retry behavior, that requires a separate architecture decision and proof.

11. Relationship to External UTC Clock Evidence

The External UTC Clock Evidence Boundary v1 is newer than the Phase 3 proof and remains intentionally independent. `VerifiedExternalClockEvidence` is time evidence only. Any future use of externally verified time inside `Provider03`, `Bind`, current rechecks, or execution freshness requires a separate explicit composition contract.

12. Phase 4 Deliverables

1. This Invariant Review Specification.
2. CAGE-side review and confirmation from Lars.
3. Resolution of the `action_context` collision rule.
4. Invariant status recorded as PASS / OPEN / NOT APPLICABLE.
5. Optional machine-readable invariant manifest if useful.
6. A Phase 5 entry decision.

13. Phase 4 Exit Criteria

Phase 4 is complete only when the invariant set is reviewed from the CAGE side, no unresolved invariant permits authority/admissibility amplification, `ESCALATE` remains non-admitted and human-review-bound, fail-closed behavior stays monotonic, `hash/evidence/TrustLog/routing` domains stay distinct, the `action_context` collision rule is explicit, and Phase 5 scope can be stated without weakening the frozen VERITAS execution path.

14. Phase 5 Entry Gate

Phase 5 — Runtime Prototype may begin only after Phase 4 completion. The runtime prototype is a new proof surface; Phase 4 approval does not itself prove live interoperability, production safety, customer readiness, or operational deployment.

15. Completion Wording

VERITAS / CAGE Phase 4 Invariant Review COMPLETE.

The `Provider03` interoperability boundary invariants have been reviewed and fixed as entry constraints for Phase 5 Runtime Prototype work.

This does not claim live production integration, customer deployment, regulatory approval, Google product adoption, or commercial integration.

16. Final Review Summary

Phase 4 exists to prevent a successful Phase 3 proof from being weakened during runtime integration.

Interoperability may add connectivity.

It must not add authority.

Any runtime prototype that cannot preserve these invariants must stop before external-effect wiring.